As a developer is writing changes to the code, the changes can be uploaded into the branch that developer is working on. This process of uploading changes is called a **commit**. However, those changes typically are not merged directly into the production environment. They must be tested and evaluated by other people and by automated testing systems. Once fully approved, then the changes to the branch can be merged into the production environment's code.

When one or more proposed changes are incorporated into the production environment, a new version of that deployment is created, documented, and stored. This information can be helpful if changes need to be reversed, or if past changes need to be referenced, analyzed, or audited.

Additionally, multiple branches can run concurrently. For example, one team might be working on one part of the code, and another team on another part. Version control software, such as Git or GitHub, is used to manage these various branches. Typically, development work is sufficiently defined so teams aren't working on the same part of the code at the same time. However, sometimes conflicts happen. For example, two developers might make different changes to the same code, or one developer might delete a file that another developer was editing. Version control systems can help identify these conflicts and add conflict markers to prevent merging until the conflicts have been resolved.

Organizations with multiple people working on the same systems will institute guidelines and best practices to help minimize these conflicts. For example, developers should commit their changes often to the branch they're working on so that small changes are incorporated before conflicts occur. Developers should perform some of their own testing to identify problems before they commit their changes. Furthermore, frequent and thorough communication between teams ensures everyone is working together toward common goals.

Self-Check

9. What is a system design characteristic that is increased by IaC?

a. Playbook
b. Drift
c. Multitenancy
d. Agility

10. What resource can be used to create a VM in the cloud, install web server software on it, and deploy a firewall to manage traffic to the VM?

a. Central repository
b. Template
c. Branch
d. Commit

☐ Check your answers at the end of this chapter.

Summary

Section 7-1: Physical Architecture

- The overall network design—the devices involved, how they're configured, the services implemented to support the network, and the way devices are connected to the network—is called a network's architecture.
- An unmanaged switch provides plug-and-play simplicity with minimal configuration options and has no IP address assigned to it. Managed switches, on the other hand, can be configured via a command-line interface or a web-based management GUI, and sometimes can be configured in groups.
- In a MAC flooding attack, the attacker sends large numbers of messages to the switch with different MAC addresses, which fills up the table and forces the switch to broadcast all messages to all ports. The Cisco command (which is also used on Arista devices) to secure switch access ports is `switchport port-security`. This is essentially a MAC filtering function that also protects against MAC flooding, which makes it a type of flood guard.

- Redundancy allows data the option of traveling through more than one switch toward its destination, and it makes your network less vulnerable to hardware malfunctions. A potential problem with redundancy has to do with traffic loops. STP (Spanning Tree Protocol) and similar technologies eliminate or otherwise control switching loops to greatly reduce the potential for a broadcast storm.
- Cisco and other manufacturers have developed a hierarchical design for switches on a network. The three tiers are the access layer (edge layer), the distribution layer (aggregation layer), and the core layer. A spine-and-leaf architecture (also called a collapsed core architecture) collapses the core and distribution layers into one layer called the spine.
- In the world of networking, the term "availability" refers to how consistently and reliably a connection, system, or other network resource can be accessed by authorized personnel. It's often expressed as a percentage, such as 98% or 99.5%. There's a difference between availability (the ability to access a resource) and durability (the resource's ongoing existence). Similar terms include "reliability," which refers to how well a resource functions without errors, and "resiliency," which refers to a resource's ability to recover from errors even if it becomes unavailable during the outage.
- The key to fault tolerance in network design is supplying multiple paths that data can use to travel from any one point to another. Link aggregation is the seamless combination of multiple network interfaces or ports to act as one logical interface, and it can help solve problems like network bottlenecks. More sophisticated load balancing for all types of servers can be achieved using a load balancer, which is a device dedicated to the task of distributing traffic intelligently among multiple devices so that every device carries a portion of the load.

Section 7-2: Virtual Architecture

- Virtualization is a virtual, or logical, version of something rather than the actual, or physical, version. A hypervisor creates and manages a VM, and it allocates hardware resources for the host and any of its guest VMs. A type 1 hypervisor installs on a computer before any OS and is, therefore, erroneously called a "bare-metal" hypervisor. A type 2 hypervisor installs in a host OS as an application and is called a hosted hypervisor.
- The way a vNIC is configured determines whether the VM is joined to a virtual network or attempts to join the physical LAN that the host machine is connected to. These various configurations are called networking modes, the most common of which are bridged, NAT, and host-only.
- Virtualization offers several advantages, including efficient use of resources; cost and energy savings; fault and threat isolation; and simple backups, recovery, and replication. Potential disadvantages to creating multiple guests on a single host machine include compromised performance, increased complexity, increased licensing costs, and a single point of failure.
- Merging physical and virtual network architecture is called NFV (network functions virtualization). NFV provides flexible, cost-saving options for many types of network devices, including virtual servers, data storage, load balancers, and firewalls.
- SDN (software-defined networking) is a centralized approach to networking that removes most of the decision-making power from network devices and instead handles that responsibility at a software level. To do this, SDN separates into pieces all the functions of a system into three layers, or planes: infrastructure plane (data plane), control plane, and application plane.

Section 7-3: Cloud Architecture

- "Cloud computing" refers to the flexible provision of data storage, applications, or services to clients over the Internet. All cloud services have the following features in common, according to NIST (National Institute of Standards and Technology): on-demand self-service, broad network access, resource pooling, measured service, and rapid elasticity.
- Cloud service models incrementally increase the amount of management responsibilities outsourced to cloud vendors. Common cloud service models include IaaS (Infrastructure as a Service), PaaS (Platform as a Service), SaaS (Software as a Service), and XaaS (Anything as a Service).
- Cloud services are delivered in a variety of deployment models, depending on who manages the underlying hardware and who has access to it. The main deployment models you are likely to encounter are public cloud, private cloud, community cloud, hybrid cloud, and multicloud.
- Generally, cloud services are categorized according to their purpose. Popular service categories include compute, networking, storage, and security.

Section 7-4: Automating Networks

- An infrastructure that is constantly adapting to changes, input, and other criteria is called a dynamic inventory. The process of using text-based commands in a computer-readable configuration file to create and manage cloud resources is called IaC (infrastructure as code).
- A programmed, computer-generated response to a specific event is referred to as automation. However, automation is limited to specific, single events. Many cloud management tasks can be automated to work together in a complex and lengthy workflow, which is called orchestration.

- As infrastructure becomes more dependent on code to run, adapt, and maintain accountability, admins and architects must use systems traditionally designed for software development to track infrastructure changes. A central repository is a storage space that contains authoritative versions of the code, especially the version that is currently deployed in the production environment. Organizations with multiple people working on the same systems will institute guidelines and best practices to help minimize conflicts between branches of code that are being developed.

Key Terms

For definitions of key terms, check the Glossary.

access layer
aggregation layer
agility
application aware networking
application plane
architecture
availability
branch
bridged mode
broadcast storm
central repository
cloud computing
cloud native
cloud service model
clustering
collapsed core architecture
commit
community cloud
container
control plane
core layer
CSP (cloud service provider)
data plane
defense in depth
distribution layer
durability
dynamic inventory
east-west traffic
edge layer
elasticity
EoR (end of row) switching
failure
fault
fault tolerance
guest
HA (high availability)
host
host-only mode
hybrid cloud
hypervisor
IaaS (Infrastructure as a Service)
IaC (infrastructure as code)
infrastructure plane
LACP (Link Aggregation Control Protocol)
layer 3 switch
layer 4 switch
least cost path
link aggregation
load balancer
load balancing
MAC flooding attack
managed switch
management plane
multicloud
multitenancy
NAT (Network Address Translation) mode
NFV (network functions virtualization)
NIC teaming
north-south traffic
on-premises data center
orchestration
PaaS (Platform as a Service)
pipeline
playbook
port aggregation
private cloud
production environment
public cloud
reliability
resiliency
root bridge
root port
SaaS (Software as a Service)
SDN (software-defined networking)
SDN controller
serverless compute
shared responsibility model
source control
spine-and-leaf architecture
STP (Spanning Tree Protocol)
three-tiered architecture
ToR (top of rack) switching
type 1 hypervisor
type 2 hypervisor
unmanaged switch
uptime
VIP (virtual IP address)
virtualization
vNIC (virtual NIC)
vSwitch (virtual switch)

Acronyms Checklist

The acronyms in Table 7-2 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 7-2 Chapter 7 acronyms

Acronym	Spelled out
CPU	central processing unit
IaaS	Infrastructure as a Service
IaC	infrastructure as code
LACP	Link Aggregation Control Protocol
NFV	network functions virtualization
PaaS	Platform as a Service
RSTP	Rapid Spanning Tree Protocol
SaaS	Software as a Service
SDN	software-defined networking
VIP	virtual IP address

Review Questions

1. What virtual, logically defined device operates primarily at the data link layer to pass frames between nodes?
   a. Virtual firewall
   b. Virtual switch
   c. Virtual router
   d. Virtual load balancer
2. With which network connection type does the VM obtain IP addressing information from the physical network?
   a. Bridged mode
   b. Host-only mode
   c. NAT mode
   d. Isolation mode
3. Which type of switch connects all devices in one rack to the rest of the network?
   a. ToR switch
   b. Spine switch
   c. EoR switch
   d. Core switch
4. What information does the `switchport port-security` command use to restrict access to a switch's interface?
   a. Port number
   b. IP address
   c. MAC address
   d. Broadcast address
5. Which cloud management technique executes a series of tasks in a workflow?
   a. Automation
   b. YAML
   c. SLA
   d. Orchestration
6. What are two advantages to using virtualization on a network?
   a. Increased complexity
   b. Efficient use of resources
   c. Single point of failure
   d. Fault and threat isolation
7. Which resource can ensure repeatability of tasks?
   a. Code branch
   b. Repository
   c. Playbook
   d. Pipeline
8. What kind of hypervisor is installed directly on top of the server's firmware?
   a. Type 1 hypervisor
   b. Bare metal hypervisor
   c. Type 2 hypervisor
   d. Hosted hypervisor
9. On a network using STP, there can be only one _____.
   a. broadcast storm
   b. least cost path
   c. blocked port
   d. root bridge
10. What protocol is most often used to bond ports between a switch and a busy server?
   a. RSTP
   b. LACP
   c. STP
   d. VIP

Scenario-Based Questions

Question 7-1

You're setting up a cluster of web servers connected by a switch to a database server. To ensure redundancy, you decide to aggregate two links from the database server's two network interfaces to the switch. In addition to availability, what other performance characteristic will this link aggregation accomplish?

a. Increased latency
b. Increased speed
c. Increased security
d. Increased bandwidth

Question 7-2

You've configured a cluster of VMs to run your customer's website, and you've connected the VMs to a redundant pair of database servers on the backend. You've selected a load balancing algorithm for your load balancer, configured security rules to filter traffic, and established an authentication directory. To which of these many devices will you attach the public VIP?

a. Authentication directory
b. Database servers
c. Load balancer
d. Web server cluster

Question 7-3

You've been tasked with providing infrastructure to a team of app developers who are working on a new system for HR to track overtime and comp days. The app will be accessed by employees on their mobile phones, tablets, and laptops. The developer team needs to test the app in several OS environments, and they want to run the code directly in the cloud without managing servers. Which type of cloud service will meet all these requirements?

a. IaaS
b. PaaS
c. Storage
d. Compute

Question 7-4

You work as a website developer on a team designing some improvements to your company's internal website. As you finish your to-do list for the morning, which consists of several minor changes to a portion of the website's search engine code, a coworker asks if you'd like to join a small group for lunch. What should you do before you leave?

a. Merge your changes.
b. Commit your changes.
c. Branch your changes.
d. Reverse your changes.

Hands-On Projects

Note 7-6 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 7-1: Access the AWS Management Console

Estimated time: 30 minutes

Objective 3.5: Compare and contrast network access and management methods.

Resources:

- AWS account (instructions for free accounts and free credits are included in the following steps)
- Internet access

Description:

Note 7-7 | **To Instructors:** AWS Academy offers a plethora of helpful and free resources for schools, instructors, and students. To begin, a central point of contact (CPOC) for your school (typically a faculty member) must apply as an institution to AWS Academy. If you're not sure whether your school is already an AWS Academy member institution, you can check the list of institutions, which is linked on the AWS Academy webpage. Be sure to submit the application a few weeks before student access is needed to allow time for response to your application, setting up your course, and exploring the resources.

After the institution is accepted, the CPOC invites faculty members to join AWS Academy. Once joined, the instructor can create a course in the AWS Academy LMS. For the projects in this book, the Academy Learner Lab course provides the required resources, specifically, access to the Learner Lab environment. The Learner Lab environment gives students a sandboxed environment in AWS where they can use their free credits to create cloud resources. The projects in this book provide the needed instructions for each project once students have entered the Learner Lab environment.

At the time of this writing, students can only join AWS Academy when you send an email invite from the AWS Academy course. Again, this is a free resource for students, and it does not require a credit card from students, faculty, or the institution. The Learner Lab course provides students with free credits, and you have access to their AWS environments so you can monitor or help troubleshoot their work in AWS.

For more information, visit aws.amazon.com/training/awsacademy. If instructors have questions or need assistance, you can contact AWS Academy staff or email the author at jillwestauthor@gmail.com.

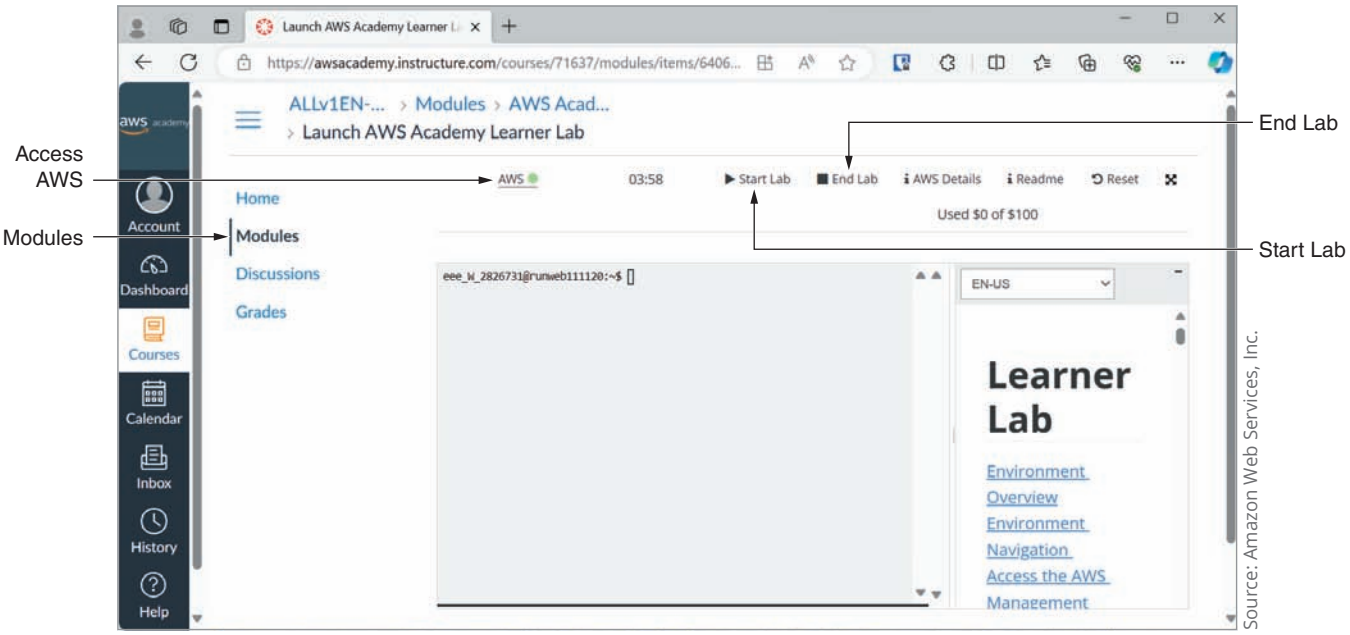
The Hands-on Projects in this chapter use the AWS (Amazon Web Services) public cloud platform. The following steps can help you create an AWS account if you don't have one already. Complete the following steps:

1. If you don't already have an AWS account, you'll need to create one. Choose one of the following options:
   a. **AWS Academy**—Your instructor might have an AWS Academy course for you. In this case, your instructor should send you an invitation email to join the course. Follow the steps given in the email. No credit card is required.

b. **Independent account**–Alternatively, you can create a standard, free account with AWS directly. When you first create an AWS account, you get some Always Free services and 12 months of Free Tier services that allow you to test-drive some services within predefined limits. The AWS projects in this course can be completed within the limits of Free Tier services at the time of this writing, although this could change. Be sure to read and understand the terms and conditions of Free Tier services, which services are included, and what are the defined limits. To sign up, go to **aws.amazon.com/free/**. A credit card *is* required for this option.

2. After creating your AWS account, regardless of the approach you used, save your sign-in information in your Bitwarden vault.
3. Sign into your AWS management console.
   a. If you're using a standard AWS account (*not* AWS Academy), sign in directly at **aws.amazon.com**.
   b. If you're using an AWS Academy course, you'll need to access your AWS management console through your AWS Academy portal: Access the course, go to the **Modules** page, and find the **Learner Lab** link. (At the time of this writing, the link is called **Launch AWS Academy Learner Lab**.) Start the lab environment, wait for it to be generated (the AWS button will turn green, the arrows will stop circling, and a command prompt will appear), and then click the **AWS** button to enter AWS (Figure 7-43).

Figure 7-43 Start your lab environment



Source: Amazon Web Services, Inc.

4. Most AWS resources reside in a specific geographical Region in the world. Regular AWS accounts have access to nearly all these Regions. However, AWS Academy accounts are limited. The Region is listed in the top right corner of your console. Click the arrow to see all the available Regions. What Region is currently selected in your account?
5. At the top of the console, click **Services**. Recall that many public cloud platforms group their services by categories, such as Compute, Networking, and Storage. What are three categories that look interesting to you?

6. Click the **Networking & Content Delivery** category. Scroll down and click **VPC**. On the VPC dashboard, you have access to specific types of resources by Region, such as VPCs, subnets, route tables, and NAT gateways. Which of these resource types do you think would let you set DNS server addresses for your VM instances in AWS?
7. The VPC dashboard lists how many of each resource type exists in your account within the currently selected Region. **Take a screenshot** of the number of VPC resources listed under Resources by Region; submit this visual with your responses to this project's questions.
8. In the left pane, scroll through the long list of services and features available in the VPC dashboard. What are three services you think are used to secure VM instances in AWS VPCs?
9. So far, you have not yet been instructed to create any resources in AWS that could accrue charges or deplete your free credits. In Project 7-2, you will create a VM instance in AWS EC2 (Elastic Cloud Compute). You can proceed with that project now or sign out of your accounts and sign back in when you're ready to complete that project.

Project 7-2: Create a Cloud VM Instance in AWS

Estimated time: 30 minutes
Objective 1.3: Summarize cloud concepts and connectivity options.
Resources:
- AWS account created in Project 7-1
- Internet access

Description:
This project picks up where Project 7-1 left off. In this project, you create and then connect an EC2 instance, which is essentially a VM in AWS. First, you will make some selections to configure the instance. Once the instance is up and running, you will connect to it so you can interact with it. To connect to an EC2 instance, AWS offers an option to connect to the instance from within your browser. This feature works for Linux instances but not for Windows instances.

Caution | Depending on the status of your account and the selections you make during this project, an EC2 instance and its supporting resources (such as storage) can deplete your credits or accrue charges. Make sure to follow these steps carefully and delete all created resources at the end of the AWS projects in this chapter.

Complete the following steps:

1. If necessary, sign into your AWS account.
   a. If you're using a standard AWS account (*not* AWS Academy), sign in directly at aws.amazon.com.
   b. If you're using an AWS Academy course, you'll need to access your AWS management console through your AWS Academy portal: Access the course, go to the **Modules** page, and find the **Learner Lab** link. Start the lab environment, wait for it to be generated, and then click the **AWS** button to enter AWS.
2. At the top of the console, click **Services**. Click the **Compute** category, and then click **EC2**. EC2 (Elastic Compute Cloud) is one of the oldest and most used AWS services. The EC2 dashboard shows you how many of each EC2 resource type you have in your account, which should currently be 0 across the board except for one security group and one key pair. Scroll down and click the **Launch instance** button.
3. Give your instance a name.

4. Most of the default selections are already correct for this project. Confirm or change to the following selections:

Caution | If you are using a standard AWS account (not an AWS Academy account) for these projects, be careful to choose "Free tier eligible" options wherever possible.

   a. Amazon Machine Image: **Amazon Linux** (use the most recent version)
   b. Instance type: **t2.micro**
   c. Key pair: **Proceed without a key pair**
   d. Firewall: **Allow SSH traffic from Anywhere 0.0.0.0/0**
   e. Configure storage: 1x **8** GiB **gp3** Root volume

Note 7-8 | A GiB (gibibyte) is similar to a GB (gigabyte), but mathematically they are not the same. A GB is typically calculated as a power of 10 (10^9 or 1,000,000,000 bytes). A GiB is calculated as a power of 2 (2^{30} or 1,073,741,824 bytes). Many cloud platforms use GiB for certain metrics.

5. Click **Launch instance**.
6. Scroll down and click **View all instances**. At first, the instance is listed as Running and Initializing.
7. Select the instance and scroll down to see additional details about the instance. Your instance was automatically assigned a public IPv4 address. If you had a web server running on the instance, this is the IP address you would assign to your URL. Your instance also has a private IPv4 address for communication within its own subnet. **Take a screenshot** of the Instance summary information section that shows the public and private IPv4 addresses; submit this visual with your responses to this project's questions.

With this information in hand, you're ready to connect to your EC2 instance:

8. With the instance selected, click **Actions** and click **Connect**.
9. Notice you have four connection options. What information is required to use the SSH client?
10. On the EC2 Instance Connect tab, make sure the connection type is set to **Connect using EC2 Instance Connect**. Then click the **Connect** button.
11. When the connection is established, enter the command `ip addr`. Which interface is assigned the private IPv4 address you found earlier when examining the instance's information?
12. Interact with your Linux instance using at least three of the Linux networking commands you learned in earlier chapters. What commands did you practice using?
13. Create a directory named `mydir` and create a new file in the directory. Enter the command `ls -R` to show the directory and the file you created.
14. **Take a screenshot** of the output; submit this visual with your responses to this project's questions.

Note 7-9 | If you wanted to remote into a Windows EC2 instance, you would need to allow RDP (Remote Desktop Protocol) traffic through the instance's firewall, and then connect to the instance using PuTTY (which is an application you install on your local computer so it can remote into a Windows computer).

When you're finished working with your instance, you're ready to power down the machine and delete the instance:

15. To shut down the machine, enter the command `sudo shutdown -h now`. Wait a minute or two as the connection is severed, and then close the connection's browser tab.
16. Refresh the data in your console for your instances list. What state is your instance in now?
17. Turning the instance off does not delete any resources in your AWS cloud. In some cases, you continue to be charged for resources even if they're turned off. To delete your instance, select the instance, click **Instance state**, and click **Terminate instance**. Click **Terminate**.
18. Refresh the instance list. What is the instance state now?
19. Sign out of your AWS account. If you're using AWS Academy, return to the AWS Academy course and end the lab environment.

Caution | Depending on the status of your account and the selections you made during the AWS projects, your EC2 instance and its supporting resources (such as storage) can deplete your credits or accrue charges. Double-check to make sure you've terminated all resources you created in the AWS projects.

Capstone Projects

Note 7-10 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 7-1: Secure Switch Ports in Packet Tracer

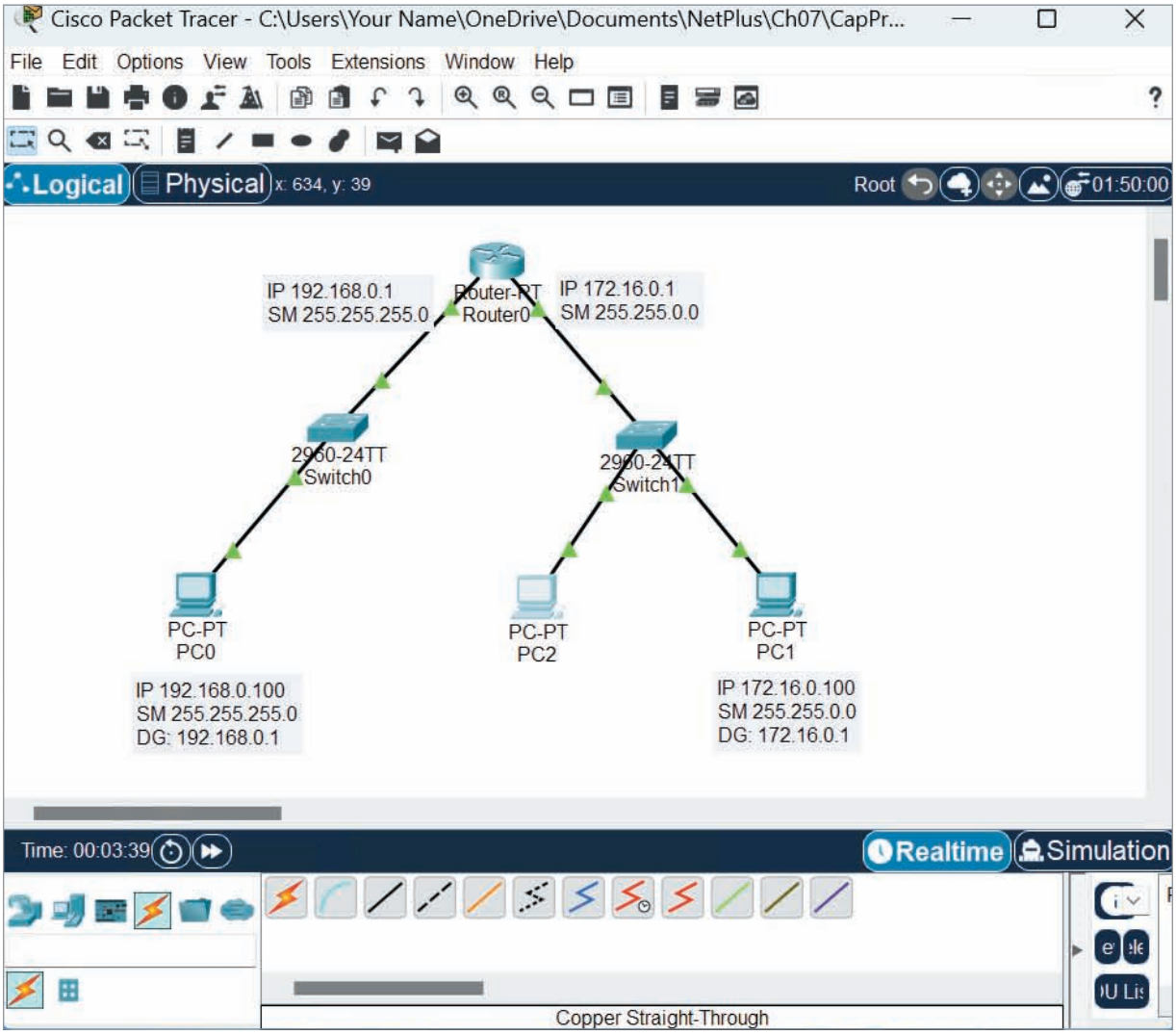
Estimated time: 30 minutes
Objective 5.2: Given a scenario, troubleshoot common cabling and physical interface issues.
Resources:
- Computer with Cisco Packet Tracer installed
- Access to the Packet Tracer network created in Capstone Project 5-1
- Storage space for Packet Tracer network file to be accessed in later chapters

Description:
In Capstone Project 5-1, you created the initial version of a network in Packet Tracer that you will continue to build on throughout other chapters. In this project, you'll add another PC to this network and explore options for securing a switch's port. You might want to save a copy of your network at this point in case you need to start over during this project. Be sure to save your network at the end of this project for use in future projects. Complete the following steps:

1. In **Cisco Packet Tracer**, open your Packet Tracer file from Capstone Project 5-1. If desired, make a backup copy before proceeding.

2. Add a new PC to your network. Create the needed connection with a Copper Straight-Through cable, connecting PC2 to Switch1's FastEthernet0/20 interface (Figure 7-44). The link between the switch and the PC should come up automatically.

Figure 7-44 Add a new PC to your Packet Tracer network



Source: Cisco Systems, Inc.

3. Configure the PC with the following information, and be sure to document your changes in the workspace:
   a. IPv4 address: **172.16.0.110**
   b. Subnet mask: **255.255.0.0**
   c. Default gateway: **172.16.0.1**
4. Use the `switchport` command to explore the options available for port security on the switch's port that faces PC2. On the switch's CLI tab, click in the IOS Command Line Interface box, press **Enter**, and then enter the commands from Table 7-3. In Step 5, there are questions for you to respond to as you work through the commands in the table, so be sure to look ahead at those questions as you work through the commands.

Table 7-3 Explore port security options on a switch

Command	Purpose
`enable`	Enters privileged EXEC mode
`configure terminal`	Enters global configuration mode
`interface fastethernet0/20`	Enters interface configuration mode for FastEthernet0/20
`switchport port-security ?`	Shows options for the `switchport port-security` command*
`switchport port-security mac-address ?`	Shows options for the `switchport port-security mac-address` command*
`switchport port-security maximum ?`	Shows options for the `switchport port-security maximum` command*
`switchport port-security violation ?`	Shows options for the `switchport port-security violation` command*

*Respond to questions in Step 5.

5. Respond to the following questions:
   a. What are the four options listed for the `switchport port-security` command?
   b. What are the two configuration options for the `switchport port-security mac-address` command?
   c. How many MAC addresses can be allowed using the `switchport port-security maximum` command?
   d. What are the three configuration options for the `switchport port-security violation` command?

Note 7-11 | When reentering the same or similar commands repeatedly at the IOS CLI, you can press the up arrow on your keyboard to recall recent commands. Then make edits to a recalled command before entering it.

Now you're ready to configure port security on the switch's port that faces PC2. To do this, you must first configure the port for access mode. This indicates to the switch the port will be connected to an endpoint rather than to another switch. Complete the following steps:

6. On the switch's CLI tab, enter the commands from Table 7-4.

Table 7-4 Configure port security on a switch

Command	Purpose
`switchport mode access`	Sets access mode for this port
`switchport access vlan 1`	Assigns this port to VLAN 1 (you'll learn more about VLANs later)
`switchport port-security`	Enables port security on the port
`switchport port-security maximum 1`	Sets the maximum number of MAC addresses allowed on this port (default value is 1)

Continues

Table 7-4 Configure port security on a switch (*continued*)

Command	Purpose
`switchport port-security violation shutdown`	Requires the port to shut down if port-security is violated (default is shutdown)
`switchport port-security mac-address sticky`	Instructs the port to dynamically learn the MAC addresses of connected devices until the maximum number is reached, and to remember those MAC addresses, that is, to make them "stick"
`exit`	Returns to global configuration mode
`exit` and press **Enter**	Returns to privileged EXEC mode
`copy run start` and press **Enter**	Saves the current settings

7. Now that you've secured the port and instructed the switch to automatically learn and remember the MAC address of the connected computer, you're ready to check the MAC address table. Enter the command **`show mac-address-table`**. What MAC address information is listed?
8. The switch has not yet seen any packets from PC2 to learn its MAC address. So you need to send some packets across the interface from which the switch can collect MAC address information. From PC2, ping PC1. What command did you use?
9. On Switch1, rerun the **`show mac-address-table`** command and respond the following questions:
   a. Now what information is listed?
   b. What is the type of each entry?
   c. Why do you think this is?
10. Pretend a hacker gains physical access to the Ethernet cable connected to PC2. They remove the cable from PC2 and connect it to their own laptop. What will happen when they try to access the network? To find out, do the following:
    a. Delete the cable connecting PC2 to the switch as follows: Click anywhere in the workspace to deselect any objects, click the **Delete (Del)** button in the Common tools bar, click the cable leading to PC2, and then press **Esc** to exit Delete mode. What information is reported on the switch's CLI tab?
    b. Add a laptop to the workspace. Set its static IP address information to the following:
       - IPv4 address: **172.16.0.110**
       - Subnet mask: **255.255.0.0**
       - Default gateway: **172.16.0.1**
    c. Add a Copper Straight-Through connection from the laptop to the switch's FastEthernet0/20 port. Wait for the link to come up. What information is reported on the switch's CLI tab now?
11. The connection is successfully established, but can the hacker do anything on the network? After both ends of the new connection turn to green triangles, go to the laptop's Command Prompt interface. Position the laptop's configuration window off to the side of the workspace so you can monitor the network and enter commands at the same time. Ping PC1, watch the effect, and then respond to the following questions:
    a. What happened when you attempted the ping?
    b. What information is reported on the switch's CLI tab?
    c. What does "administratively down" mean?

12. The switch has automatically shut down the port in response to the security violation. **Take a screenshot** of your network showing the administratively down link; submit this visual with your responses to this project's questions.
13. Now try to reconnect the legitimate computer. Delete the laptop and its connection to the switch, and add a new connection between PC2 and the switch's FastEthernet0/20 port. What happens?
14. Try to ping PC1 from PC2. What happens?
15. To reenable the connection from PC2, go to the switch's CLI tab. Position the switch's configuration window off to the side of the workspace so you can monitor the network and enter commands at the same time. Press **Enter** to return to the command prompt, and then enter the commands from Table 7-5. What happens to the connection?
16. After both ends of the new connection turn to green triangles, ping PC1 from PC2 to confirm the connection is restored. Save your project in a safe place for future projects.
17. Make some notes on your Wikidot website about your activities in Packet Tracer for this project.

Table 7-5 Restart a switch's port

Command	Purpose
`configure terminal`	Enters global configuration mode
`interface fastethernet0/20`	Enters interface configuration mode for FastEthernet0/20
`shutdown`	Disables the port
`no shutdown` and press **Enter**	Enables the port

Note to instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

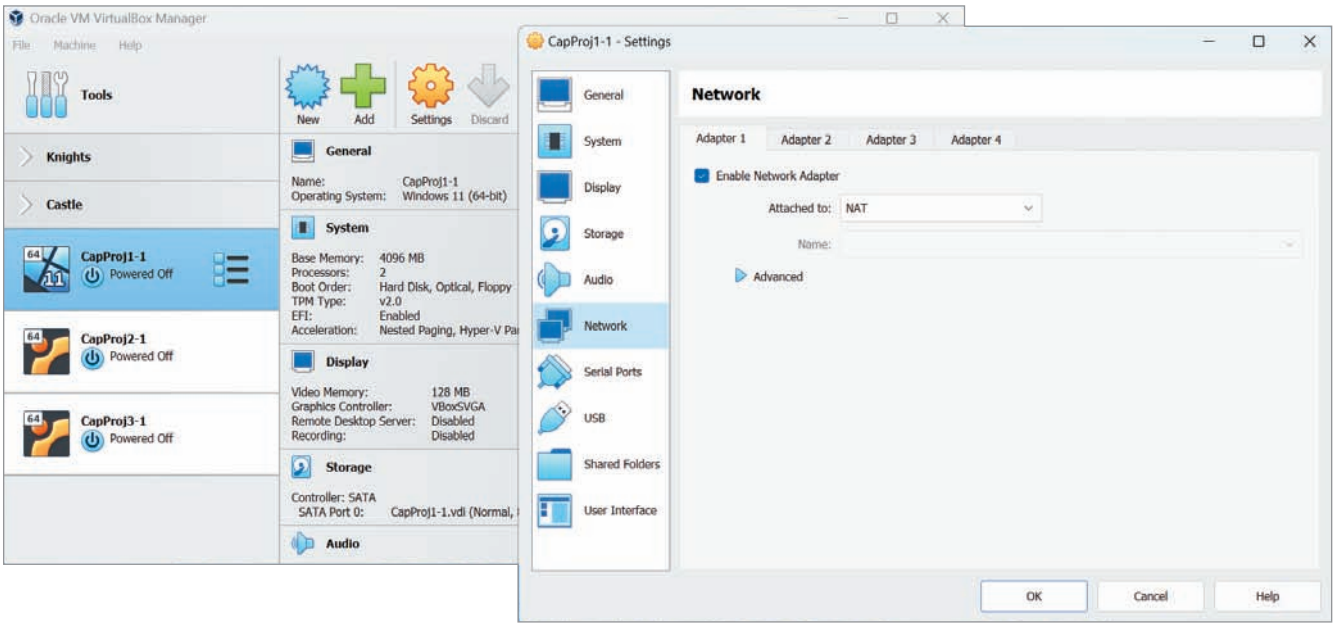
Capstone Project 7-2: Explore Virtual Network Configuration Options in VirtualBox

Estimated time: 30 minutes
Objective 1.2: Compare and contrast networking appliances, applications, and functions.
Resources:
- Access to the same computer used to complete Capstone Project 1-1
- Internet access

Description:
In the Chapter 1 Capstone Project, you created a virtual machine using Oracle VirtualBox. In this project, you will explore the network settings for your VM in VirtualBox, practice communicating between the VM and the host machine, and practice communicating between the VM and a host on the Internet. The host computer, which is the physical computer, should be connected to the Internet. Complete the following steps:

1. Open **VirtualBox**. To ensure everyone starts this project from the same configuration, select your Windows VM and click **Settings**. In the left pane, click **Network**. Make sure the **Adapter 1** tab is selected (Figure 7-45). If necessary, change the *Attached to* option to **NAT**. Click **OK**. Start your **Windows VM**.

Figure 7-45 The NAT option is selected



Source: Oracle Corporation

2. In the VM, open a PowerShell or Command Prompt window and enter `ipconfig`.
3. **Take a screenshot** of the command output; submit this visual with your responses to this project's questions.
4. Check the VM's connection to the Internet by pinging one of Google's DNS servers with the command `ping 8.8.8.8`. Was the ping successful?
5. Open a PowerShell or Command Prompt window on the host computer, the physical PC hosting the VirtualBox VM. Enter `ipconfig`. What are the host machine's IPv4 address, subnet mask, and default gateway for the active, physical network connection (either Ethernet or Wi-Fi)?
6. Compare the host computer's address information with the VM's information. Both computers should have a subnet mask of 255.255.255.0, which means both computers use the first three octets to identify the network ID. Look at the first three octets of each computer's IPv4 address and respond to the following questions:
   a. Do both network IDs match? If so, they're on the same subnet. If not, the computers are on different subnets.
   b. Are these two computers on the same subnet? Why do you think this is?
   c. Ping the host PC from the guest VM. Does it work?

Note 7-12 | If you're completing this project in a school lab, your computer might be using a different subnet mask. In that case, the subnet mask could be 255.0.0.0 or 255.255.0.0. For 255.0.0.0, look at the *first* octet only to determine whether the VM and its host are on the same subnet. For 255.255.0.0, look at the first *two* octets to determine whether the VM and its host are on the same subnet. Continue with these parameters for the following steps.

7. Return to the Oracle VM VirtualBox Manager window and open the VM's **Settings** window. In the Settings window, click **Network** and make sure the **Adapter 1** tab is selected.
8. Click the **Attached to** arrow, click **Bridged Adapter**, and then click **OK**.
9. Return to the VM's window. You might see a message indicating that the VM has detected a network change, and it's requesting permission to find PCs and other devices and on the network. If so, click **Yes**.
10. Repeat the `ipconfig` command and respond to the following questions:
    a. What are the IPv4 address, subnet mask, and default gateway now?
    b. How does this information compare with the host PC's information?
    c. Are the two computers on the same subnet now?
    d. Ping the host PC from the guest VM. Does it work?
11. On the VM, ping Google's DNS server again. Was the ping successful?
12. On the Oracle VM VirtualBox Manager window, return to the VM's **Settings**, **Network** menu, click the **Attached to** arrow, click **Host-only Adapter**, and then click **OK**.
13. On the VM, repeat `ipconfig`, ping the host PC, and ping Google's DNS server again. What changed?
14. On the host PC, run `ipconfig` again and respond to the following questions:
    a. Which adapter on the host PC's output has an IPv4 address on the same subnet as the VM?
    b. What does this tell you about the host machine's physical and virtual network connections?
15. Change the VM's "Attached to" setting to **NAT** and click **OK**. On the VM, repeat `ipconfig`, ping the host PC's physical adapter, and ping Google's DNS server again. What changed?
16. Close all windows in the VM, shut down the VM, and close all windows on the host machine.

Solutions to Self-Check Questions

Section 7-1: Physical Architecture

1. Which STP bridge serves as the basis for all path calculations?
   Answer: d. Root bridge
   Explanation: The **root bridge**, or master bridge, provides the basis for all path calculations. Only one root bridge exists on a network. From this root bridge, a series of logical branches, or data paths, emanate like branches on a tree.

2. Which type of switch is best used for connections to web servers?
   Answer: a. Edge switch
   Explanation: In a three-tiered architecture, **edge switches** connect directly to hosts such as servers, printers, and workstations. Ideally, hosts connect *only* to edge (also called access) switches and never to switches at the other layers, although this is not always the case in the real world. Aggregation switches, or distribution switches, provide a highly redundant mesh of connections that route traffic within the corporate network as well as providing traffic filtering and the network's connection to a WAN. The core layer consists of highly efficient multilayer switches or routers that support the network's backbone traffic. In a two-tiered spine-and-leaf architecture, spine switches on the backbone connect in a mesh topology with all leaf switches, which in turn connect with servers and other host devices.

3. What is a group of links called in Windows?
Answer: b. NIC teaming
Explanation: NIC teaming on Windows devices is the seamless combination of multiple network interfaces or ports to act as one logical interface. Fault tolerance is the capacity of a system to continue performing despite an unexpected hardware or software malfunction. Load balancing is a distribution of traffic over multiple components or links to optimize performance and fault tolerance. Clustering refers to the technique of grouping redundant resources such as servers so they appear as a single device to the rest of the network.

Section 7-2: Virtual Architecture

4. Which virtual network connection type assigns a VM its IP address from the physical network?
Answer: b. Bridged
Explanation: Although a **bridged** vNIC communicates through the host's adapter, it obtains its own IP address, default gateway, and subnet mask from a DHCP server on the physical LAN. In NAT mode, the VM obtains IP addressing information from its host rather than from a server or router on the physical network. In host-only mode, VMs use the DHCP service in the host's virtualization software to obtain IP address assignments. In Hyper-V, the host-only connection type is created by assigning VMs to a private virtual network.

5. Which network architecture technique are you using when you run a virtual router on a network?
Answer: c. NFV
Explanation: NFV (Network Functions Virtualization) is a network architecture that merges physical and virtual network devices. STP (Spanning Tree Protocol) is a switching protocol defined by the IEEE standard 802.1D that prevents traffic loops by artificially blocking the links that would complete a loop. A VIP (virtual IP address) is a single IP address that represents a group of devices in a cluster. SDN (software-defined networking) is a centralized approach to networking that removes most of the decision-making power from network devices and instead handles that responsibility at a software level.

6. Which SDN plane moves traffic from switch port to switch port?
Answer: d. Infrastructure plane
Explanation: The **infrastructure plane** is made up of the physical or virtual devices that handle actual transmissions on the network, moving traffic from each port on a switch to other ports. The control plane handles the decision-making processes performed by network protocols. The application plane includes applications running on the network, such as DNS or VoIP. The management plane could be considered a part of the control plane and allows network administrators to remotely manage network devices, monitor those devices, and analyze data collected about the devices.

Section 7-3: Cloud Architecture

7. Which cloud characteristic ensures you can manage cloud resources from an iPad?
Answer: d. Broad network access
Explanation: Broad network access means client devices of all types can access services, applications, and storage in a cloud, no matter what operating system they run or where they are located, if they have an Internet connection. Rapid elasticity means resources can be changed quickly and dynamically—even automatically—in response to changing demands. Resource pooling refers to the fact that host computers in the cloud provide multiple services or resources such as disk space, applications, and services that are pooled, or consolidated, usually among multiple tenants. Cloud-native refers to a design approach that incorporates services built to best take advantage of the cloud environment.

8. When you set private IP address ranges for servers in your cloud, what service model are you using?
Answer: a. IaaS
Explanation: IaaS (Infrastructure as a Service) provides virtual hardware services, including network infrastructure devices such as virtual servers and DNS services. PaaS (Platform as a Service) provides a platform on which to test or run code without having to manage the underlying server. SaaS (Software as a Service) consists of applications provided through an online user interface that are compatible with a variety of devices and operating systems. DaaS (Desktop as a Service) is a type of SaaS where you can access a virtual desktop through your browser. With PaaS, SaaS, and DaaS, cloud customers typically cannot configure networking structures such as subnets and IP address ranges.

Section 7-4: Orchestration and Automation

9. What is a system design characteristic that is increased by IaC?
Answer: d. Agility
Explanation: IaC (infrastructure as code) increases a system's ability to adapt quickly to market demands according to increased or decreased feature use within the system, which is referred to as **agility**. A playbook is a collection of tasks to be orchestrated in a specific sequence and that are defined in a template. Unauthorized, unpredicted, or undocumented changes made to a system can cause configuration drift, which is the gradual departure from approved system configurations. Multitenancy is a characteristic of cloud computing where several customers (tenants) pay for resources running on the same hardware.

10. What resource can be used to create a VM in the cloud, install web server software on it, and deploy a firewall to manage traffic to the VM?
Answer: b. Template
Explanation: A collection of tasks to be orchestrated in a specific sequence are defined in a **template**, also called a playbook. A central repository is a storage space that contains authoritative versions of code. Each time a new copy is made for development and testing, a branch is created. This process of uploading changes from a developer to a branch is called a commit.